Human Development Index (HDI) Predictor
Problem Solution Fit

Project: Human Development Index (HDI) Predictor
Phase: Project Design Phase

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1. The Problem

Estimating a country's human development level from raw indicators requires manually applying the UNDP's geometric-mean formula across three sub-indices, which is slow and error-prone for quick comparisons or what-if scenarios.

2. Why a Regression Model Fits

The four chosen indicators are strongly correlated with HDI — correlation coefficients of 0.83 to 0.97 in the project's own exploratory data analysis. That strength of relationship means a simple, interpretable Linear Regression model captures the mapping with high accuracy, while staying easy to explain to non-technical stakeholders such as policymakers or students.

3. Fit Validation

- High correlation between inputs and target confirms a linear model is a reasonable choice
- Held-out test R^2 of approximately 0.99 confirms the fit generalizes well
- Predictions were verified at both ends of the development spectrum (very high and low HDI countries)

4. Why Not a More Complex Model

Given the strong linear correlation already present between indicators and HDI, a more complex model (e.g. gradient boosting or a neural network) would add training and deployment overhead without a meaningful accuracy gain, while sacrificing the interpretability that matters for this use case.